LearnHub

Lecture notes: by date and by subject (minor differences may occur between the two versions)

Class exercises and their solutions

Reading assignments:

- Reading assignment 1: Chapter 1 (Assessments) due next Monday.
- You will have 7-10 days to work on each reading assignment.

Labs:

- Lab 3 today, upload only PDF
- Lab 2 grades are on LearnHub

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Agenda

Point estimators

- Bias
- Variance
- Mean square error

Maximum likelihood estimators

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Point estimators (Section 4.9)

Let $X_1,\ldots,X_n$ be a (simple) random sample.

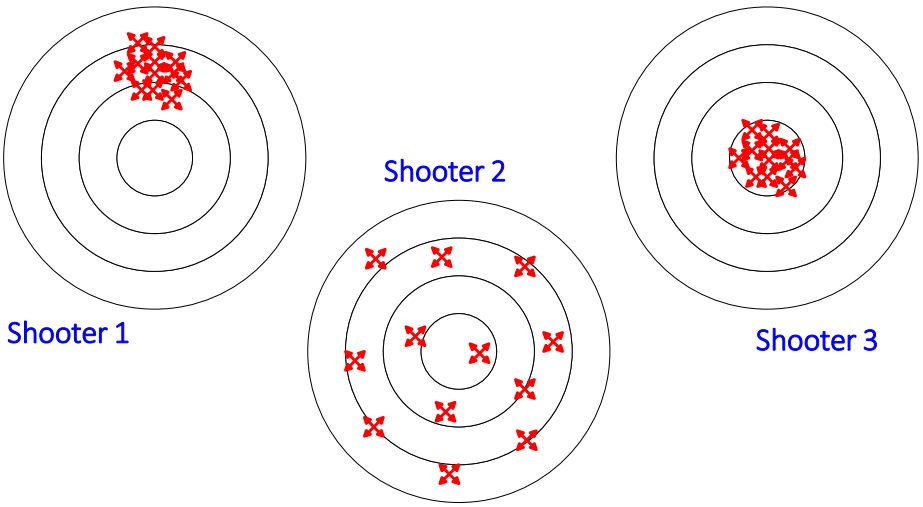
A statistic that is used to estimate the value of a parameter is called a point estimator.

A statistic is any function of the observations in a random sample.

A statistic is a random variable, so it has its own probability distribution (generally called the sampling distribution).

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Which shooter would you hire if you were the chief officer of the police station?



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Point Estimation

Target → Population parameter θ
(such as mean, variance, median etc.)

Shot → Point estimate of a parameter θ
(a sensible value for θ, a single number)

Shooter → Point estimator of a parameter θ
(a rule, mostly a formula, to calculate an estimate based on the sample)

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Point Estimation

Let X be a random variable with probability distribution f(x; θ) where θ is an unknown parameter.

Let $X_1, X_2, \ldots, X_n$ be a random sample from X.

$\Rightarrow$ the statistic $= h(X_1, X_2, \ldots, X_n)$ corresponding to θ is the estimator of θ.

Note: The estimator is itself a random variable since it is based on sample.

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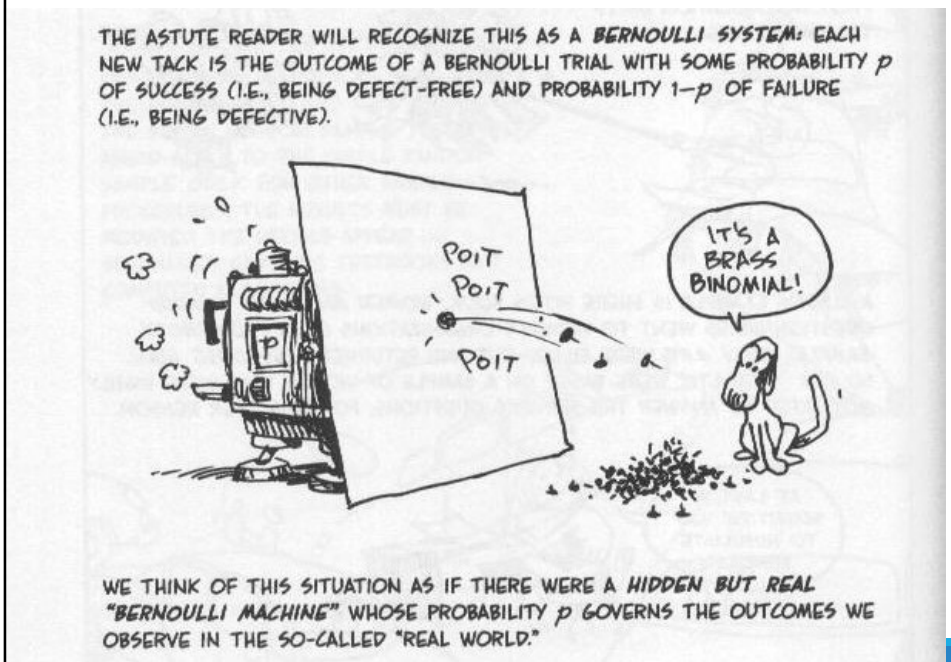
Simple questions

$X_1, X_2, \ldots, X_n$ be a random sample from the random variable X.

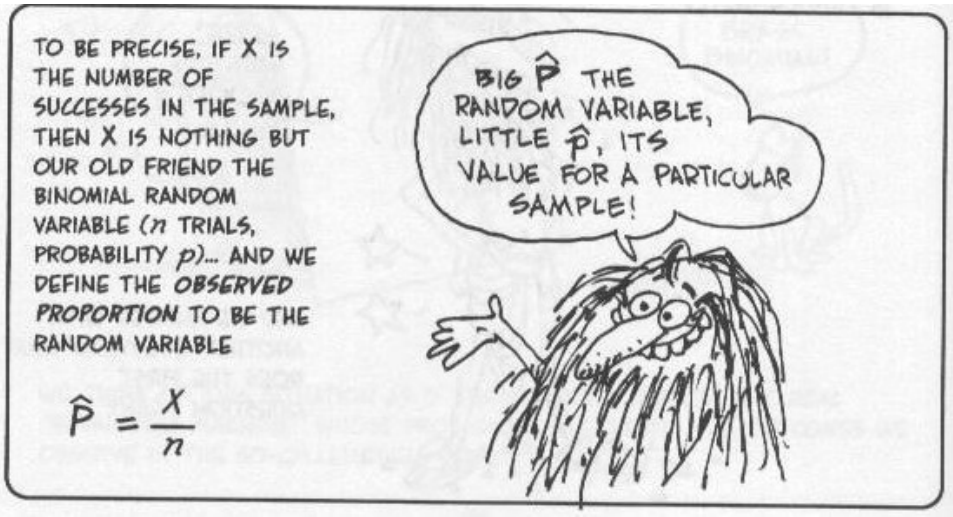
- How to estimate the mean of X from this sample?
- How to estimate the standard deviation of X from this sample?
- How to estimate the median of X from this sample?

Other simple questions…

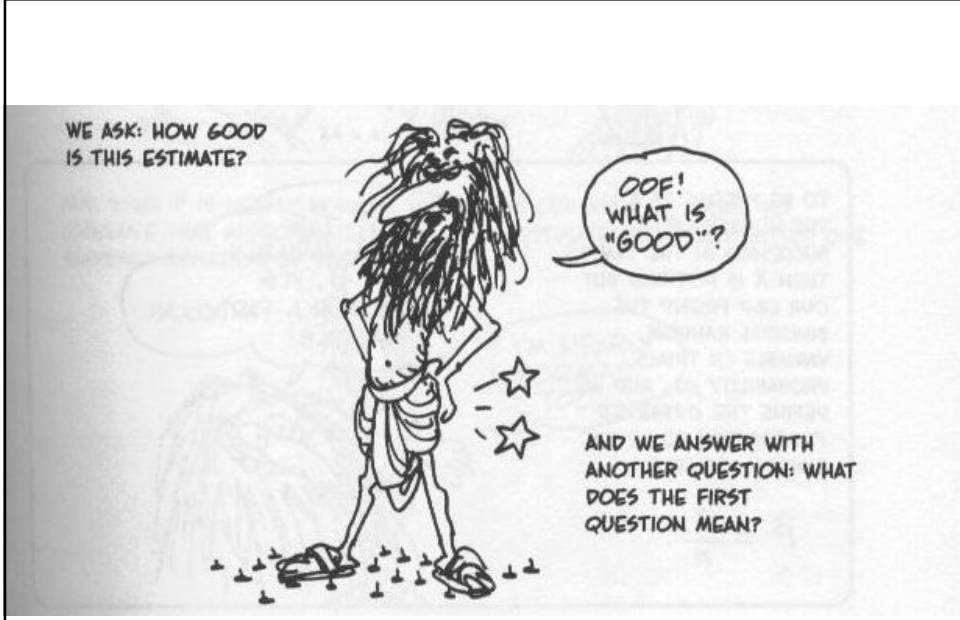
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TO BE PRECISE, IF X IS THE NUMBER OF SUCCESSES IN THE SAMPLE, THEN X IS NOTHING BUT OUR OLD FRIEND THE BINOMIAL RANDOM VARIABLE (n TRIALS, PROBABILITY p)... AND WE DEFINE THE OBSERVED PROPORTION TO BE THE RANDOM VARIABLE
$\hat{P} = \frac{X}{n}$
BIG $\hat{P}$ THE RANDOM VARIABLE, LITTLE $\hat{p}$, ITS VALUE FOR A PARTICULAR SAMPLE!

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WE ASK: HOW GOOD IS THIS ESTIMATE?
OOF! WHAT IS "GOOD"?
AND WE ANSWER WITH ANOTHER QUESTION: WHAT DOES THE FIRST QUESTION MEAN?

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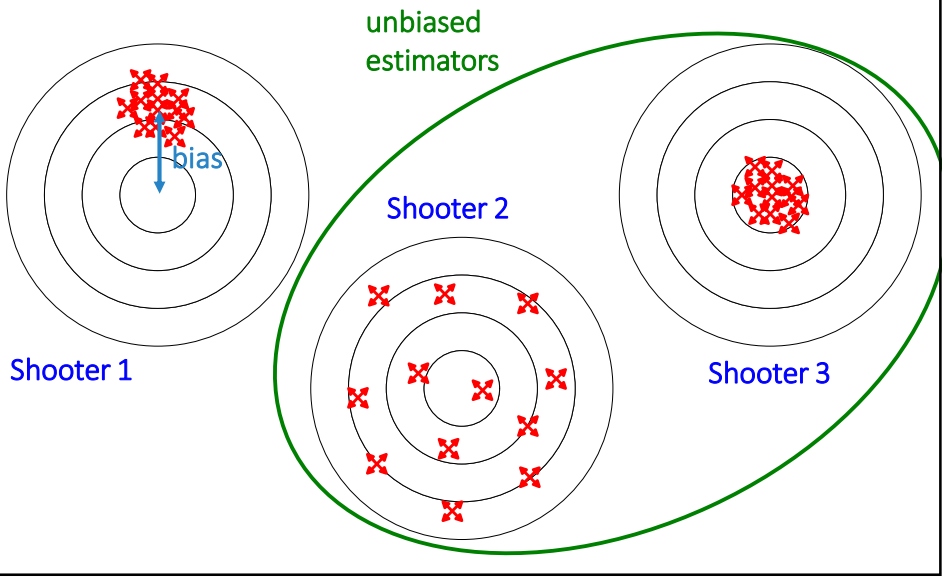
Unbiased estimators

Definition: A point estimator $\hat{\theta}$ is said to be an unbiased estimator of θ if $\mathrm{E}[\hat{\theta}] = \theta$ for all possible values of θ.

If $\hat{\theta}$ is not unbiased, $\mathrm{E}[\hat{\theta}] - \theta$ is called the bias of $\hat{\theta}$.

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Which shooter would you hire if you were the chief officer of the police station?



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Mean Square Error of an Estimator

Definition

The **mean square error** of an estimator $\hat{\Theta}$ of the parameter θ is defined as

$$\text{MSE}(\hat{\Theta}) = E(\hat{\Theta} - \theta)^2$$

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Mean Square Error of an Estimator

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$$\text{MSE}(\hat{\Theta}) = E(\hat{\Theta} - \theta)^2 \qquad (7\text{-}3)$$

$$MSE(\hat{\Theta}) = E[(\hat{\Theta} - \theta)^2] = E[(\hat{\Theta} - E[\hat{\Theta}] + E[\hat{\Theta}] - \theta)^2]$$

a b

$$= E[(\hat{\Theta} - E[\hat{\Theta}])^2 + (E[\hat{\Theta}] - \theta)^2 + 2(\hat{\Theta} - E[\hat{\Theta}])(E[\hat{\Theta}] - \theta)]$$

a^2 b^2

$$= E[(\hat{\Theta} - E[\hat{\Theta}])^2] + E[(E[\hat{\Theta}] - \theta)^2] + 2E[(\hat{\Theta} - E[\hat{\Theta}])(E[\hat{\Theta}] - \theta)]$$

$$= \underbrace{E\left[(\hat{\Theta} - E[\hat{\Theta}])^2\right]}_{\text{Variance}} + \underbrace{(E[\hat{\Theta}] - \theta)^2}_{(\text{Bias})^2} + \underbrace{2(E[\hat{\Theta}] - \theta)E[(\hat{\Theta} - E[\hat{\Theta}])]}_{0}$$

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Mean Square Error of an Estimator

$$\mathrm{MSE}_{\hat{\theta}} = \left(\mu_{\hat{\theta}} - \theta\right)^2 + \underbrace{\left(\sigma_{\hat{\theta}}\right)^2}_{\text{standard error of } \hat{\theta}}$$

Definition: $\sigma_{\hat{\theta}} = \sqrt{V(\hat{\theta})}$ is called the standard error of an estimator $\hat{\theta}$.

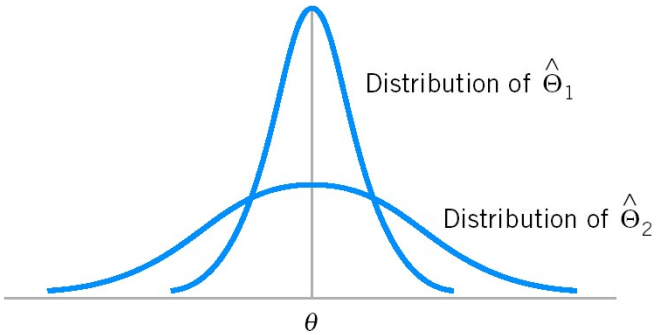
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General Concepts of Point Estimation

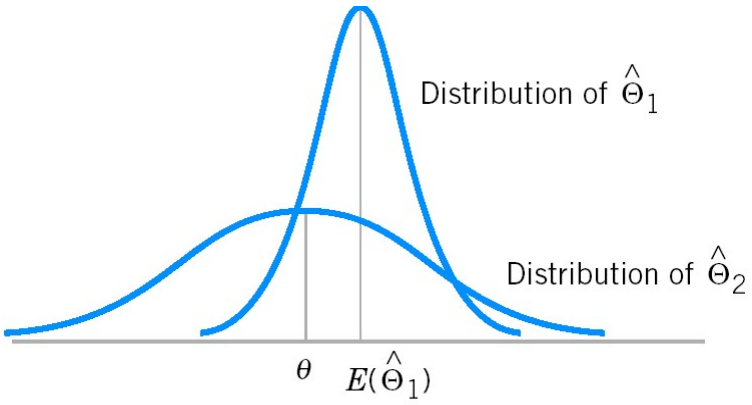
Variance of a Point Estimator

The sampling distributions of two unbiased estimators $\hat{\Theta}_1$ and $\hat{\Theta}_2$.



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Mean Square Error of an Estimator



A biased estimator $\hat{\Theta}_1$ that has smaller variance than the unbiased estimator $\hat{\Theta}_2$.

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Which estimator is unbiased?

Definition: A point estimator $\hat{\theta}$ is said to be an unbiased estimator of θ if $\mathrm{E}[\hat{\theta}] = \theta$ for all possible values of θ.

If $\hat{\theta}$ is not unbiased $\mathrm{E}[\hat{\theta}] - \theta$ is called the bias of $\hat{\theta}$.

Is $\hat{p}$ unbiased? Can we find its MSE?

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What else is unbiased?

Proposition: Let X be a random variable with mean μ & variance σ^2. Let $X_1, \ldots, X_n$ be a random sample of size *n* from X.

a. $\bar{X}$ is an unbiased estimator of μ.

MSE of $\bar{X}$: ?

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